Large-Scale 3D Reconstruction from Sparse and Noisy Walkthroughs

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Abstract

Accurate large-scale 3D reconstruction of active indoor construction sites from monocular video remains challenging due to repetitive structures, dynamic workers, motion blur, and long unbounded trajectories. We present a complete pipeline that adapts Depth Anything 3 (DA3) [5]—a feed-forward transformer that jointly estimates dense depth and camera poses from unposed images—to 360° panoramic video recorded on the HILTI×Trimble Challenge 2026 dataset. Our contributions include IMU-guided equirectangular-to-pinhole projection with four-yaw tiling, Grounded-SAM-2 dynamic-object confidence masking, a fixed device-occlusion mask, yaw-group pose filtering, and an extended loop-closure window for long-range trajectory correction. Evaluated across 30 construction-site sequences, our system achieves full pose coverage on all runs and, on the five sequences with ground-truth trajectories, obtains a mean XY RMSE of 1.075 m and a proxy XY score of 72.5. Code is available at `https://github.com/TangentH/da3-hilti`.

1. Introduction

Tracking the progress of a construction project requires regular, accurate 3D measurements of the site. Traditional photogrammetry and laser scanning deliver centimetre-level accuracy but demand specialist hardware and careful setup. A compelling alternative is to use ordinary walkthroughs recorded on a 360° camera worn by a site worker—but this introduces a different set of challenges: the camera moves irregularly, the environment changes between visits (workers, scaffolding, equipment), lighting is often poor, and identical corridors and rooms create perceptual aliasing that confuses place-recognition systems.

Classical Structure-from-Motion (SfM) and SLAM pipelines [10] rely on handcrafted feature matching and iterative bundle adjustment, both of which can fail catastrophically when features are sparse or repeated. Feed-forward reconstruction methods [5, 11, 12] have recently shown that a single transformer trained end-to-end can predict globally consistent camera poses and dense depth maps in a single forward pass, bypassing iterative refinement entirely. Their robustness to low-texture scenes and their ability to run on unposed image collections make them an attractive fit for the construction-site scenario.

We build on Depth Anything 3 (DA3) [5] and its streaming extension (DA3-Streaming), which processes arbitrarily long image sequences using a sliding-window chunked inference scheme with loop-closure correction. Our work adapts this system to the specific constraints of the HILTI×Trimble Challenge 2026 dataset [2]: 360° equirectangular panoramas captured simultaneously with an IMU, long trajectories that revisit floors over many months, and persistent dynamic objects (construction workers and their helmets).

Contributions. The following adaptations and implementation choices extend the upstream DA3 and DA3-Streaming codebases for this dataset:

- **IMU-guided four-yaw pinhole projection.** We read IMU accelerometer data from the ROS2 bag, apply a first-order low-pass filter to estimate the gravity direction, and use the resulting rotation to project each equirectangular panorama into four axis-aligned 768×512 pinhole views (yaw $0°, 90°, 180°, 270°$) with the horizon leveled. This converts the panoramic input into a format natively handled by DA3 without any fine-tuning.
- **Grounded-SAM-2 confidence-zero masking.** We detect and segment workers and helmets in every pinhole view using Grounded-SAM-2 [9] and suppress the corresponding depth predictions by setting their confidence to zero, preventing dynamic objects from corrupting the point cloud.
- **Static device-occlusion mask.** The 360° camera hardware is partially visible in the equirectangular panoramas. We build a single binary mask of the device footprint and project it into each of the four yaw views, ensuring the camera body is never back-projected into the 3D reconstruction.
- **Yaw-group pose filtering.** Because every physical camera location yields exactly four consecutive frames (one

per yaw angle), we exploit this four-frame grouping to detect and discard point-cloud chunks whose estimated poses are internally inconsistent (intra-group spread $> 0.10\,\mathrm{m}$), which reliably flags hallucinated structures.

- **Extended loop-closure window.** The default DA3-Streaming loop-closure parameters are tuned for short sequences. We expand the minimum frame gap, the top-k retrieval count, and the SALAD [3] similarity threshold to match the long, revisiting trajectories of multi-floor construction sites.

2. Related Work

Feed-forward 3D reconstruction. DUSt3R [12] introduced the paradigm of regressing pixel-aligned 3D point maps from unposed image pairs, eliminating the need for explicit feature matching or bundle adjustment. MASt3R [4] extended this with a dense matching head that improves both reconstruction quality and pose accuracy. VGGT [11] scaled feed-forward reconstruction to larger image sets using a global attention backbone. Depth Anything 3 [5] unifies monocular depth, multi-view depth, and camera pose estimation under a single depth-ray representation built on a plain DINOv2 [7] encoder with DualDPT decoding head, and outperforms VGGT on most benchmarks while keeping the architecture minimal. MonST3R [13] extends the DUSt3R framework to dynamic scenes by estimating per-frame scene flow alongside geometry. To the best of our knowledge, our work is among the first systematic applications of DA3-Streaming to sparse 360° helmet-camera walkthroughs from active construction sites. Rather than proposing a new feed-forward reconstruction model, we focus on the application-specific adaptations required to make DA3-Streaming usable in this setting.

Dynamic-object removal. Removing transient objects such as pedestrians from 3D reconstructions is a well-studied problem. Recent segmentation models like SAM 2 [8] can propagate masks temporally in a zero-shot manner. Grounding them with open-vocabulary detectors such as GroundingDINO [6] avoids the need for manually annotated training data for each object class. Grounded-SAM-2 [9] combines both components into a single inference pipeline. We adopt Grounded-SAM-2 with a text prompt tuned to construction-site occupants (`person.helmet.`) and use the resulting masks to suppress depth confidence rather than inpainting the image, which integrates cleanly with the confidence-aware DA3 architecture.

Visual place recognition and loop closure. NetVLAD [1] demonstrated that aggregating local CNN features into a compact VLAD descriptor is highly effective for image-based place retrieval. SALAD [3] improves this aggregation via optimal-transport soft assignment and achieves state-of-the-art place-recognition recall. DA3-Streaming integrates SALAD as its loop-closure detector, using it to retrieve candidate revisit frames before running Sim(3) alignment. Our contribution is in the *configuration* of this component: we increase the minimum temporal gap, top-k count, and similarity threshold to handle long indoor trajectories where revisits occur hundreds of frames apart.

Construction-site 3D mapping. In the HILTI×Trimble Challenge 2026 [2] setting used in this work, the sensor platform is a 360° helmet-mounted camera with an IMU. Compared with more controlled hand-held capture, this setup introduces sparse and noisy walkthrough trajectories, rapid head rotations, frequent occlusions, and strong motion irregularities, making large-scale reconstruction from visual input particularly difficult.

3. Method

Our pipeline (Fig. 1) consists of three stages: input preprocessing, dynamic-object masking, and 3D reconstruction with refinement. It is designed to run end-to-end from a single ROS2 SQLite bag to a colored point cloud, camera poses, and a fly-through video.

3.1. Input Preprocessing

Frame extraction. The HILTI camera hardware publishes 360° equirectangular panoramas to a ROS2 topic. We decode the bag with a stride of 10 (one frame per 10 published), yielding approximately 3–8 frames per second depending on the publication rate. The Kalibr calibration file provides the extrinsic transformation between the camera and IMU frames needed for gravity leveling.

IMU-guided horizon leveling. Each equirectangular panorama is projected into four axis-aligned pinhole views. Before projection we remove the camera's roll and pitch using the IMU accelerometer. Specifically, we read the `/imu/data_raw` topic from the ROS2 bag and apply an exponential low-pass filter to the raw accelerometer vector:

$$\hat{\mathbf{g}}_t = (1-\alpha)\,\hat{\mathbf{g}}_{t-1} + \alpha\,\mathbf{a}_t, \quad \alpha = 1 - e^{-\Delta t/\tau}, \tag{1}$$

with time constant $\tau = 0.25\,\mathrm{s}$. The gravity direction in camera coordinates is obtained via the pre-calibrated IMU-to-camera rotation $\mathbf{R}_{\mathrm{cam}\leftarrow\mathrm{imu}}$. We then construct an orthonormal rotation that maps camera *down* onto the gravity direction while keeping the nominal forward axis:

$$\mathbf{R}_{\mathrm{level}} = [\mathbf{r},\ \hat{\mathbf{g}},\ \mathbf{f}] \in \mathrm{SO}(3), \tag{2}$$

where $\mathbf{f}$ is the gravity-orthogonal component of the nominal forward axis and $\mathbf{r} = \hat{\mathbf{g}} \times \mathbf{f}$.

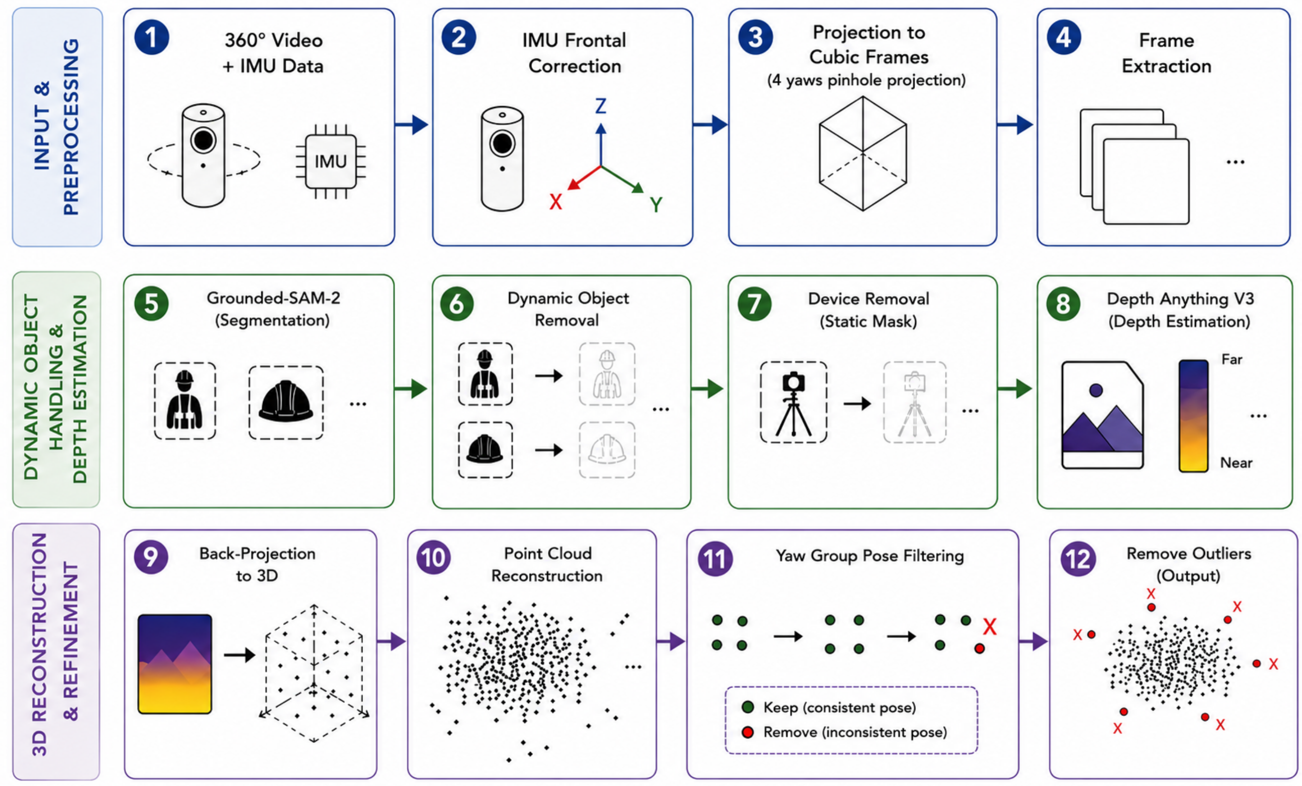

Figure 1. Overview of our pipeline. **Input & Preprocessing** (steps 1–4): ROS2 bags are decoded into 360° equirectangular panoramas, IMU measurements level the horizon, and each panorama is converted into four yaw-aligned pinhole views before frame extraction. **Dynamic Object Handling & Depth Estimation** (steps 5–8): Grounded-SAM-2 detects workers and helmets, while a projected static device mask marks camera-body artifacts; these dynamic and static masks are passed to DA3 as confidence-zero regions, so DA3 still predicts depth for the full image but masked pixels are excluded from reliable point-cloud fusion. **3D Reconstruction & Refinement** (steps 9–12): the predicted depths are back-projected into 3D and accumulated into a point cloud; yaw-group pose filtering removes locally inconsistent pose estimates, and point-cloud outlier cleaning produces the final refined static 3D scene.

Four-yaw pinhole projection. For each frame we produce four pinhole images by applying $\mathbf{R}_{\text{level}}\,\mathbf{R}_z(\phi_i)$, $\phi_i \in \{0°, 90°, 180°, 270°\}$, and remapping from the equirectangular domain using bilinear interpolation with horizontal wrapping. Output resolution is 768×512 at a $90°$ horizontal FOV ($f_x = 384\,\text{px}$). Each sequence of four images from the same timestamp forms a *yaw group* and is given sequential frame indices $[4k, 4k+1, 4k+2, 4k+3]$ for yaw $[0, 90, 180, 270]°$.

3.2. Dynamic-Object Masking

Grounded-SAM-2 masks. For every pinhole frame we run Grounded-SAM-2 [9]: GroundingDINO [6] detects bounding boxes for the text prompt `person.helmet.` (box and text thresholds both 0.3), and SAM 2 [8] segments the interiors. Binary masks are morphologically cleaned with a 3×3 closing followed by a 5×5 dilation. Each mask is saved as a `.npy` confidence-zero array consumed by DA3's masked inference path, which zeroes out the depth confidence of the masked pixels rather than discarding them from the attention computation.

Static device mask. The camera hardware occludes a small region near the bottom of every equirectangular panorama. We manually annotate this region once in the equirectangular domain, then project it into each of the four yaw views using the same remap as above and dilate by 3 pixels to account for projection boundary effects. The resulting per-view masks are merged with the Grounded-SAM-2 masks via logical OR.

3.3. DA3-Streaming Reconstruction

Sliding-window inference. DA3-Streaming [5] decomposes a long sequence into overlapping chunks of $C = 40$ frames with an overlap of $O = 20$ frames. Within each chunk, DA3 jointly estimates per-pixel depth maps and camera extrinsics in a shared reference frame. Adjacent chunks are stitched via Sim(3) alignment on their shared frames. We set a point-cloud confidence threshold of $0.75 \times \mathrm{conf\_mean}$ and sample 5% of confident points per chunk to control memory footprint.

Yaw-group pose filtering. Every physical camera location contributes four consecutive frames (one per yaw). These four camera centers should be nearly identical in 3D space; a spread larger than a few centimetres indicates a failed local reconstruction. After DA3 outputs per-chunk camera poses, we compute the 95th-percentile intra-group spread for all complete yaw groups in each chunk and discard chunks whose median group spread exceeds $d_{\max} = 0.10\,\mathrm{m}$. This reliably removes chunks where the transformer failed to converge, preventing ghost geometry from polluting the accumulated point cloud.

Loop closure. DA3-Streaming detects revisited locations using SALAD [3] visual-place-recognition descriptors extracted from each chunk's reference frame. For long construction-site sequences, the default minimum-gap setting is too conservative: we increase the minimum frame gap to 160 (40 physical camera locations), raise top-k retrieval to 20, and lower the similarity threshold to 0.85. When a loop is confirmed, a Sim(3) transformation is estimated from the overlapping chunk pair and propagated backward through the pose graph.

Point-cloud cleanup. The accumulated raw point cloud contains scattered outliers from noisy depth predictions. We apply a voxel-based neighbor filter: a point is retained only if its $0.05\,\mathrm{m}$ voxel contains at least 2 points and at least 3 of its 26-connected neighboring voxels are occupied. Removed points are saved separately for diagnostics.

3.4. Output

The final deliverables for each run are: a cleaned colored PLY point cloud (`reconstruction.ply`), a 4×4 camera-pose file (`camera_poses.txt`), a camera-center PLY for trajectory visualization, and an H.264 fly-through video at 720p. For sequences where a floor-plan image is available, we additionally perform scan-matching between the flattened point cloud and the architectural floor plan to produce a georeferenced overlay.

4. Experiments

4.1. Dataset

We evaluate on the **HILTI×Trimble Challenge 2026** dataset [2], using 30 walkthrough sequences recorded on an active multi-storey construction site with an Insta360 One-RS 1-Inch 360° camera and its integrated IMU. The sequences cover 10 floor levels or areas (Floors 1–7, EG, UG1, and UG2) across 8 recording dates between May and December 2025. Ground-truth camera trajectories, derived from LiDAR-based reference trajectories, are available for five sequences; the remaining 25 sequences are used for qualitative evaluation only.

Table 1. Per-run ground-truth trajectory evaluation on five sequences. Coverage (Cov.) is 100% for all runs. XY and 3D errors are in metres. Bold highlights best XY RMSE.

Run	XY error		3D error		Proxy	Sim(3)
	RMSE	p95	RMSE	p95	XY	scale
Fl.1 May 5	0.314	0.605	0.734	0.936	88.8	1.001
Fl.2 May 5	1.027	1.651	1.344	1.932	67.0	0.956
Fl.2 Oct.28 R1	**0.193**	**0.366**	**0.495**	**0.873**	**93.1**	0.990
Fl.2 Oct.28 R2	0.579	1.080	1.140	2.159	78.6	0.986
Fl.UG1 Oct.16	3.261	5.660	3.372	5.728	34.8	0.917

Table 2. Mean evaluation results over all five ground-truth runs.

Metric	Mean value
Map XY RMSE (m)	1.075
Map XY p95 (m)	1.872
Proxy XY score	72.5
Map 3D RMSE (m)	1.417
Map 3D p95 (m)	2.326
1s RPE RMSE (m)	0.322
Mean Sim(3) scale	0.970

4.2. Evaluation Metrics

We evaluate the final floorplan-registered trajectories directly in map coordinates, without using ground truth to align, rotate, or scale them. We report:

- **Map XY RMSE / p95**: RMS and 95th-percentile of per-frame 2D positional error (m) in the XY plane.
- **Proxy XY score**: Local application of the HILTI × Trimble Challenge scoring metric, $\frac{1}{N}\sum_{t=1}^{N} 100e^{-0.460517e_t}$, where e_t is the XY error in metres. Scores range from 0 to 100; higher is better.
- **Map 3D RMSE/p95**: RMS and 95th-percentile of position error including the vertical coordinate.
- **1s RPE RMSE**: RMSE of the difference between estimated and ground-truth translation over one-second intervals, measuring short-term motion accuracy.
- **Sim(3) scale**: Scale from a separate GT-optimal Sim(3) diagnostic. Values near 1.0 indicate little residual scale mismatch after floorplan registration.

4.3. Quantitative Results

Tables 1 and 2 summarize the quantitative results. Our system achieves 100% temporal coverage on all five ground-truth runs, meaning that every eligible ground-truth timestamp after the initial five-second exclusion can be matched to the estimated trajectory. The best results appear on Floor 2 October 28 Run 1, which has good lighting and moderate trajectory length, yielding an XY RMSE of only 0.193 m and a proxy XY score of 93.1. Performance degrades on the Underground Floor 1 (UG1) sequence, which contains a long corridor with very low ambient light, contributing to frequent depth estimation failures and a large accumulated drift (XY RMSE 3.261 m, proxy score 34.8). The mean Sim(3) scale of 0.970 indicates only a small residual scale correction after floorplan registration.

4.4. Qualitative Results

Figure 2 shows representative reconstructions from different floors and recording dates. Overall, the pipeline recovers the dominant indoor layout, including room boundaries, doorways, corridors, and large structural elements such as concrete pillars and staircases. The figure also highlights the main visible failure cases. In `floor_6 2025-06-18 run_1`, large holes appear in the reconstructed ceiling, where metallic panels or reflective membranes are present. The UG-floor examples are visibly sparser and noisier, with accumulated drift and local artifacts. We analyze the underlying causes of these failure cases in Sec. 5.

Figure 3 shows the best-performing sequence in Table 1. The final estimated trajectory closely follows the ground-truth path in map coordinates, with only small local deviations at turns and transitions. This visual agreement supports the low localization error reported in Table 1.

5. Discussion

Scope of adaptation. Rather than proposing a new reconstruction algorithm, our work adapts DA3-Streaming to panoramic helmet-camera video in industrial environments. The four-yaw pinhole decomposition, IMU-guided horizon leveling (Eqs. (1) and (2)), and yaw-group pose filtering (Sec. 3) are dataset-specific design choices that make the upstream model usable on the HILTI$\times$Trimble sequences. The confidence-zero masking approach—rather than inpainting—is a lightweight integration that avoids modifying the model's attention mechanism and could be re-used for DA3 applications with dynamic foreground objects.

Why four yaw views work. DA3 was trained on standard pinhole images, so we split each panorama into four perspective crops instead of feeding the equirectangular image directly. In our main 768×512 setting, each crop has a 90° horizontal FOV and an approximately 67.4° vertical FOV; the four yaw crops therefore tile 360° with essentially no same-position horizontal overlap. Reconstruction remains possible because DA3 has strong learned monocular and multi-view geometric priors, and because temporal chunks still contain overlap between views captured from nearby camera positions, even if different yaw crops from the same panorama do not overlap.

Device-mask design. The static device mask is hand-drawn and deliberately conservative: it is designed to cover the camera rig across all runs, so for some sequences it removes slightly more pixels than strictly necessary. In practice, the panoramic input still contains sufficient scene coverage, and this conservative mask does not visibly degrade reconstruction quality while substantially reducing artifacts from reconstructing the camera body itself. We did not rely on Grounded-SAM-2 [9] for this mask because GroundingDINO [6] makes the pipeline slow, especially as the number of detected instances increases. We also observed degraded segmentation quality with prompts containing more than three object terms, and prompts such as `device.mount`, did not reliably capture the exact hardware regions to remove. Future work could draw setup-specific masks for each camera configuration, or use a single validated mask once the acquisition setup is fixed.

Failure modes. The main failures in Figure 2 arise from low-light underground corridors, reflective materials, and local pose errors. UG1 and UG2 combine poor illumination, long textureless concrete corridors, and repeated wall structures, making both depth estimation and long-range trajectory correction difficult. Reflective surfaces cause view-dependent and inconsistent depth predictions; during fusion, these low-confidence observations are suppressed or removed, producing holes such as the ceiling incompleteness in `floor_6 2025-06-18 run_1`. Additional errors come from missed dynamic-object masks, over- or under-exposed regions, missed loop closures, and local DA3 pose failures. The yaw4 pose filter removes cases where one yaw view deviates from the others at the same camera location, but it only checks positional consistency: it does not explicitly verify orientation, and cannot detect failures where all four views drift together or remain internally smooth while being globally misregistered; its fixed threshold can also reject useful frames in low-texture regions.

Conservative loop closure. The SALAD similarity threshold of 0.85 is deliberately conservative to avoid spurious loop closures, which would introduce large trajectory jumps. As a result, several revisited locations—especially in perceptually aliased corridors—are not closed, leaving residual drift. A lower threshold together with a more robust pose-consistency check (e.g., geometric verification)

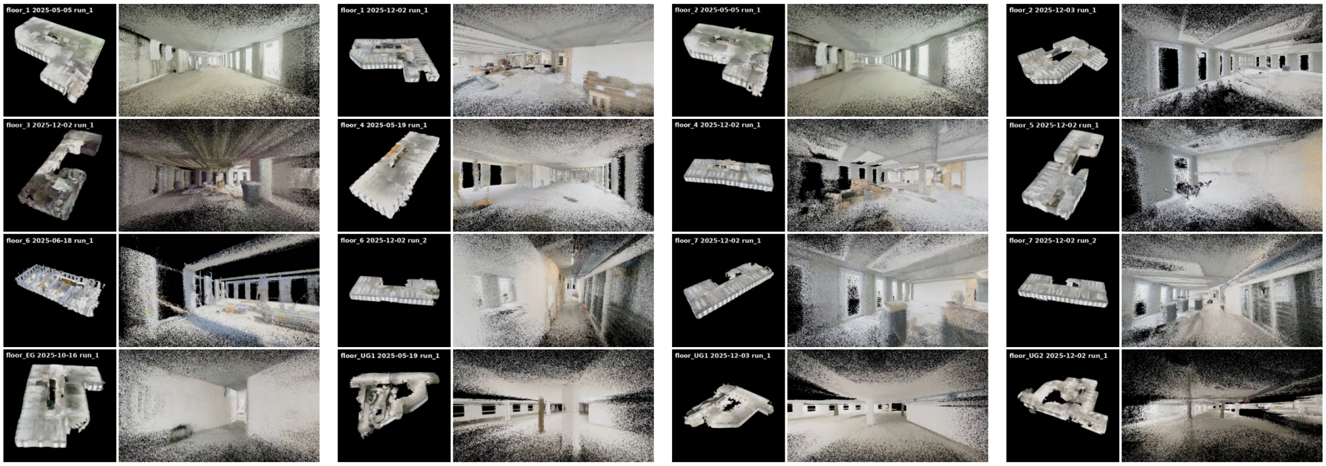

Figure 2. Qualitative reconstructions across 16 sequences from different floors and recording dates. Each example contains a bird's-eye view (BEV) of the reconstructed point cloud on the left and an interior point-cloud view on the right. The pipeline generally recovers room layouts, doorways, and structural elements. The main visible failure cases are point-cloud holes in `floor_6 2025-06-18 run_1` and sparser, noisier reconstructions with visible drift in the last three UG-floor examples.

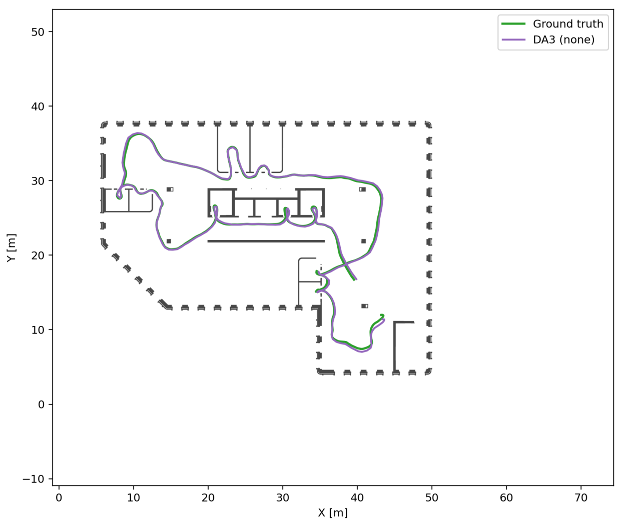

Figure 3. Trajectory overlay for Floor 2 Oct. 28 Run 1. The estimated trajectory closely overlaps the ground-truth path, qualitatively confirming the low localization error reported in Table 1.

could improve recall without increasing false-positive loop closures.

Scale accuracy. DA3 targets metric depth, while our floorplan-registration stage additionally refines the global scale. After registration, the mean GT-optimal Sim(3) scale is 0.970, indicating that the final trajectories are close to metric scale. Because this diagnostic is computed after floorplan registration, it cannot isolate the native scale accuracy of DA3.

6. Conclusion

We presented a complete pipeline for large-scale 3D reconstruction of active construction sites from sparse, noisy 360° walkthrough videos. By adapting DA3-Streaming with IMU-guided panorama decomposition, Grounded-SAM-2 dynamic-object masking, and construction-site-specific filtering, we successfully reconstructed all 30 sequences in the HILTI×Trimble Challenge 2026 dataset. The best runs achieve sub-20 cm XY RMSE and proxy scores above 90, demonstrating that feed-forward reconstruction methods can deliver practically useful 3D maps from casual walkthroughs. The primary remaining limitations—basement dark corridors and highly reflective materials—call for stronger low-light depth estimation and multi-modal sensor fusion in future work.

7. Work Distribution and Code Usage

Work distribution. All team members participated equally to the full code development, evaluation and reporting.

Code usage.
- **Used as-is:** Depth Anything 3 core model and weights [5]; DA3-Streaming sliding-window inference, Sim(3) alignment, and SALAD loop closure; Grounded-SAM-2 [9]; and the SALAD VPR descriptor [3].
- **Implemented by us:** ROS2 extraction, IMU-leveled four-yaw projection, static device masking, yaw-group pose filtering, point-cloud cleanup, floor-plan overlay, batch reconstruction scripts, result packaging, and DA3-Streaming configuration tuning.

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